23/02

=) Algorithm Binary-Search (A, n, key)

Input:

A → sorted array

n → no. of elements

key → element to be searched

Output:

position of key if found

otherwise return -1

step 1: low ← 0, high ← n-1

step 2: while low ≤ high do

mid ← (low + high)/2

step 3: if A[mid] = key then

return mid

step 4: else if key < A[mid] then

high ← mid - 1

step 5: else

low ← mid + 1

step 6: end while

step 7: return -1

TC: Best: O(1)

Avg: O(log n)

worst: O(log n)

END algorithm.